INSTITUTE
OF LAW,
POLITICS AND
DEVELOPMENT



Sant'Anna
School of Advanced Studies – Pisa

Quantitative Methods

Module I, Session 2: Confidence Intervals and Hypothesis Tests

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Section 1

1. Where we are

1.1 Last session in one slide

Identification

Would our comparison give the right answer **with unlimited data**?

→ a question about the **research design**: whose average effect, and why is the comparison fair?

Inference

How close to the right answer are we **with the data we actually have**?

→ a question about **chance**: how much would our estimate move in another sample?

Today is entirely about the right-hand column. But keep the left one in view: an interval or a test can tell you how much **chance** might be fooling you. It cannot tell you whether the **comparison** was fair in the first place.

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2. "Rich NATO members both like the US and vote with it at the UN." Which challenge to identification?
3. What is the difference between a **population parameter** and an **estimate**? Which one is fixed?
4. A distribution: what do its **centre** and **spread** tell you?

Section 2

2. Reading the agent's code

2.1 Lab 1: what did you find?

Five questions:

1. **Did it do everything it was asked?** Compare the script, step by step, with the prompt. Anything missing, anything extra?

Then: one line each in `log.md`. Who found something the agent got wrong — or *said* it did right?

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4. **Is the randomness under control?** A `set.seed()`? How many draws? With or without replacement?
5. **Do the numbers pass a sanity check** against something you know independently?

Then: one line each in `log.md`. Who found something the agent got wrong — or *said* it did right?

2.2 One script, read closely

```r
## Style follows AGENTS.md
set.seed(42)
x_large <- rnorm(n = n_large, mean = 50, sd = 10)
within_large <- mean(abs(x_large - 50) <= 10)
x_small <- rnorm(n = 20, mean = 50, sd = 10)  # x 3
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- **Good:** a seed, and `sd = 10` named: a standard deviation, not a variance. "Within 10 of 50" exactly as asked.

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- **Check:** the header says it follows `AGENTS.md`. Did it? *The agent's account of its own work is output to check.*

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- **Check:** the header says it follows `AGENTS.md`. Did it? *The agent's account of its own work is output to check.*
- **Ask:** why are the averages of 20 so much more scattered than the average of 1,000?

2.3 Five things to check in any script

Check	**Typical slip**
The whole task	One part of the prompt quietly skipped, or something added that nobody asked for
Arguments	A variance passed where R wants a standard deviation; a 28 where it wants 0.28
Rows	`na.rm = TRUE` with no word on which cases disappeared
Randomness	No seed; sampling without replacement from a small "population"
Boundaries	"More than 60" computed when the question was "60 or more"

Every one of these runs without an error message. That is the point: **code that runs is not code that is right.**

Section 3

3. Bigger samples: settling down and piling up

3.1 Bigger samples settle down

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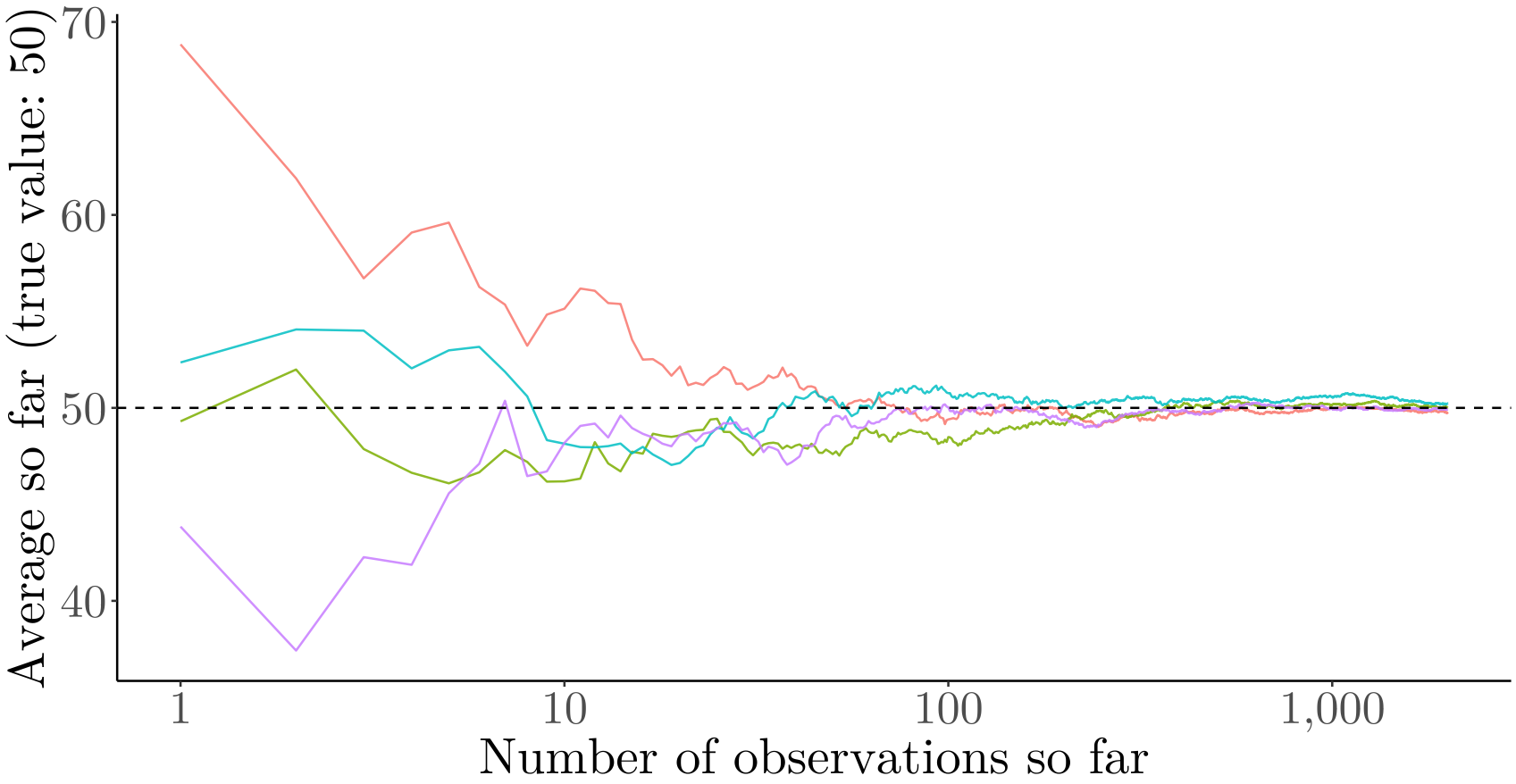
- It licenses the whole enterprise: with enough data, estimates stop wandering.
- It says nothing about *how fast*, and nothing about small samples.
- It is a statement about the **method**, not about any single sample. A large sample can still be wrong; it is just less likely to be wrong by much.

3.2 Four runs, one truth

```r
set.seed(54)
n_max <- 2000
runs <- lapply(1:4, function(i) {
  draws <- rnorm(n_max, mean = 50, sd = 10)
  data.frame(n = 1:n_max,
             m = cumsum(draws) / (1:n_max),
             run = factor(i))
})
ll <- do.call(rbind, runs)
```

What does this code do? How can we find out?

3.3 Four runs, one truth



3.4 Averages pile up in a bell curve

Knowing that averages settle down is not enough. We need to know *how* they scatter around the truth. (This is the **Central Limit Theorem**.)

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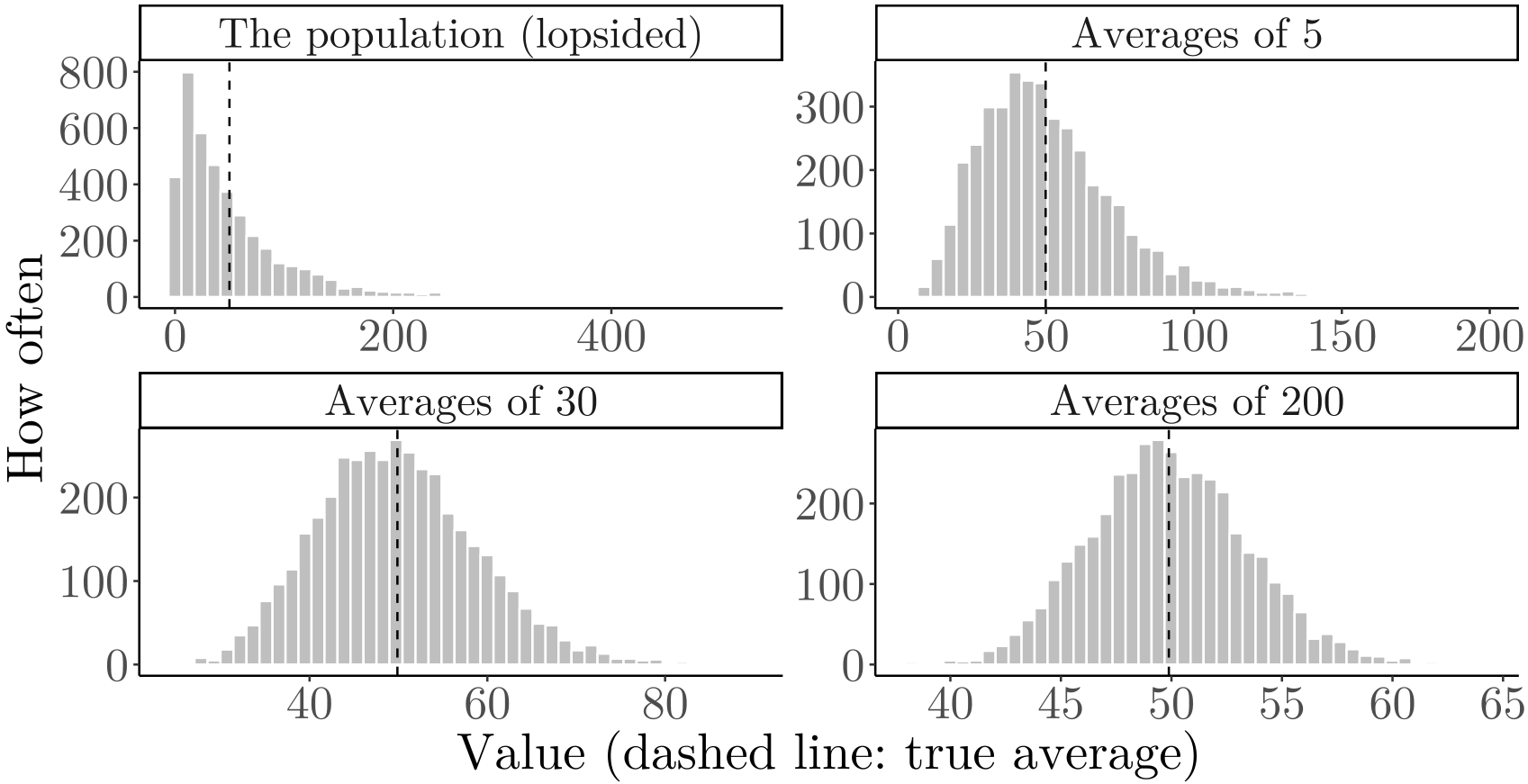
- Imagine drawing many samples of the same size and computing the average of each. Those averages pile up in a **bell curve centred on the true value**.
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- This happens **whatever the shape of the population**: lopsided, lumpy, yes/no — as long as samples are big enough.
- The spread of that pile has a name: the **standard error**. It shrinks as samples grow: **four times the data halves the error**.

3.5 Seeing it happen



3.6 A caution for your own work

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- For a roughly symmetric population, about 30 observations is already fine.
- For something like `troops_iraq` in our data — where almost every state contributes zero and one contributes 466,985 — 30 is nowhere near enough.
- Small samples and extreme values are exactly where published findings tend to break. Be suspicious of a bell-curve approximation when samples are small and the outcome is lopsided.

3.7 Why this matters for everything that follows

If we know our average comes from a bell curve **centred on the truth**, with a spread we can **estimate from our one sample**, then we can:

1. Say how far our estimate is *likely* to be from the truth — and turn that around into a range of plausible true values: a **confidence interval**.

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2. Ask how surprising our estimate would be if the true effect were zero: a **hypothesis test**.

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2. Ask how surprising our estimate would be if the true effect were zero: a **hypothesis test**.
3. Do both without ever observing the population, and without ever seeing a second sample.

Section 4

4. Standard errors

4.1 The standard error

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- In words: take how spread out the individual cases are, and **shrink it by the square root of the number of cases**.

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- Noisy subject matter pushes it up; more data pulls it down.
- For a yes/no variable the same logic holds — a share is just the average of zeros and ones. It is hardest to pin down when the split is near 50–50.

4.2 Standard deviation or standard error?

Standard deviation	How spread out the *cases* are. A description of your data.
Standard error	How spread out the *estimate* would be across samples. A statement about your uncertainty.

More data does not make states more alike, so the standard deviation does not shrink. It does make the average more precise, so the standard error does.

When you see error bars, the first question is always: bars of what?

4.3 Two surprises

- **The size of the population does not matter.** A well-drawn poll of about 1,100 people has a margin of error of roughly ± 3 points — in the United States or in Estonia (Imai 2017, ch. 7). Bigger countries do not need bigger polls.

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- **Precision costs the square of what you want.** To go from ± 3 points to ± 1 point you need not three times but about **nine times** the data: roughly 10,000 respondents.
- So underpowered studies cannot be rescued by a few more cases. Past a point, the better investment is a better design.

4.4 Standard error in practice

```
# % with a positive view of the US
x <- gh$pos1[!is.na(gh$pos1)]
n <- length(x)
mean <- mean(x)
sd <- sd(x)
se <- sd(x) / sqrt(length(x))
round(c(n=n, mean=mean, sd=sd, se=se), 1)
```

```
   n mean   sd   se
43.0 28.2 15.1  2.3
```

43 countries were surveyed. On average 28.2% held a positive view of the United States. Countries differ a lot (standard deviation 15.1 points), but the **average** is pinned down to within a couple of points (standard error 2.3).

Section 5

5. Lab: standard errors two ways

5.1 Lab A: specify, run, verify

To the agent

Write scripts/lab_se.R. Treat the observed values of unvoting in data_2011-07-26.dta as a population. Draw 2,000 samples of 10 states and 2,000 of 40; plot the averages and report their standard deviation. Compare with the formula sd/sqrt(n). Repeat for the share signing an Article 98 agreement (article98). Run it.

Verify — without asking the agent:

1. How many states have a value for `unvoting`? Did the agent say how many it dropped — and do you know *who* they are?
2. Is there a `set.seed()`? Are samples drawn with or without replacement?
3. Does four times the sample halve the spread? Compare with `sd(gh$unvoting, na.rm = TRUE) / sqrt(40)`. If they differ, explain why before calling it a bug.
4. Are the averages centred on the population average (34.6)? One line in `log.md`.

Section 6

6. Confidence intervals

6.1 Turning the bell curve around

- From the bell curve: in 95% of samples, the estimate lands within about **two standard errors** of the truth.

The truth does not move. The interval does: every new sample brings a new interval. Think of ring toss at a fair — the peg stays put, and "95%" describes how good your throwing is, not where the peg is.

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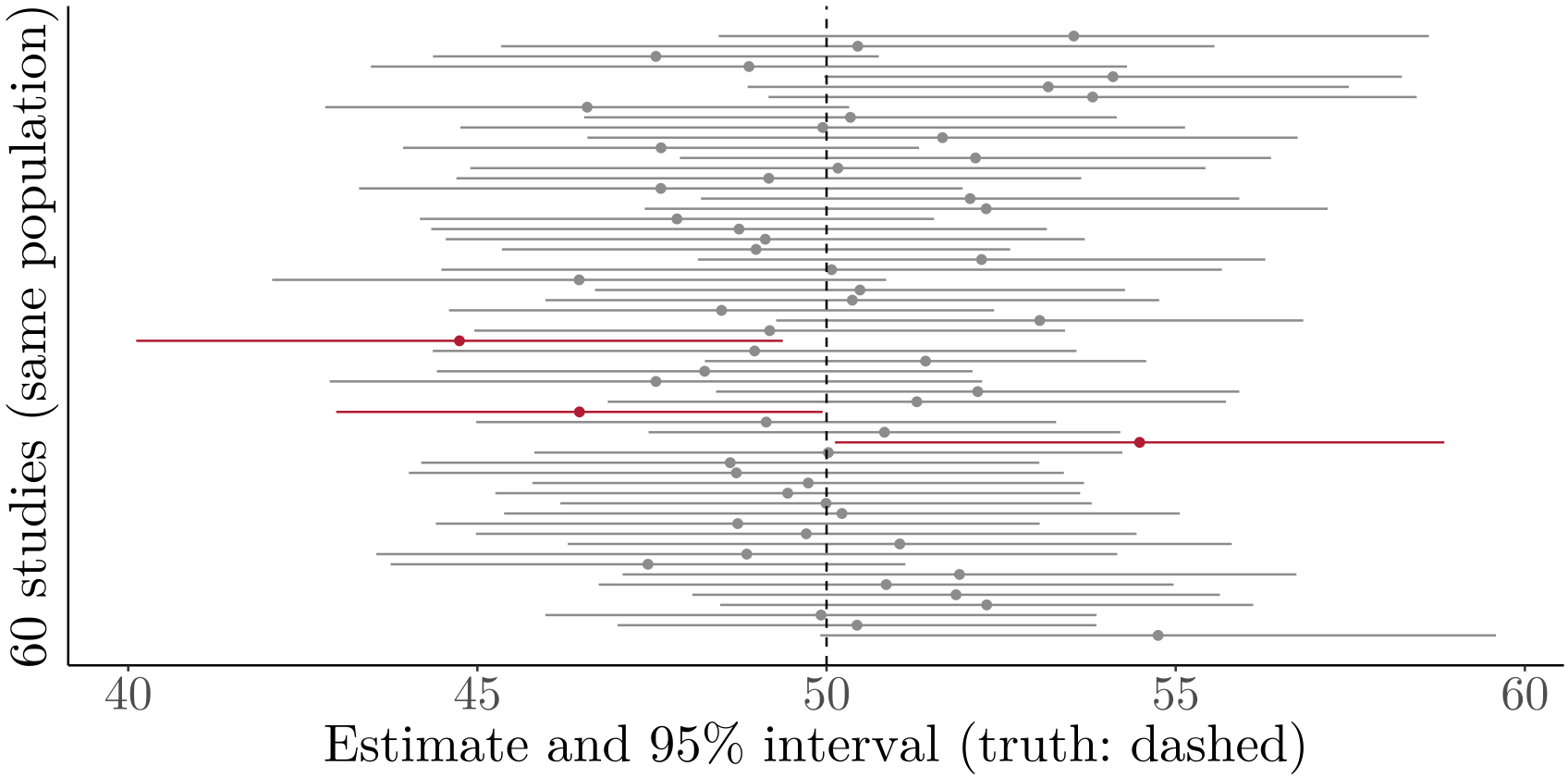
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- Turn it around: in 95% of samples, the range "estimate **plus or minus** two standard errors" **contains the truth**.
- That range is the **95% confidence interval**. Half its width is what pollsters call the **margin of error**.

The truth does not move. The interval does: every new sample brings a new interval. Think of ring toss at a fair — the peg stays put, and "95%" describes how good your throwing is, not where the peg is.

6.2 What "95 percent" looks like



6.3 How to say it

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- Right: "Intervals built this way contain the true value in 95% of samples. This is one of them."
- Textbooks write "we are 95% **confident**" (Dean & Illowsky 2018, ch. 8). That is fine — as long as you remember the confidence is in the **method**, not in this particular interval.

6.4 What makes an interval wider

If you...	**the interval...**
ask for more confidence (99% instead of 95%)	gets wider: a bigger net catches more often
collect more data	gets narrower — four times the data, half the width
study cases that differ a lot from each other	gets wider
have a very small sample	gets a little wider still (next slide)

There is no free lunch: for a given sample, more confidence means less precision.

6.5 Small samples: a slightly wider ruler

With few cases, our estimate of the spread is itself shaky. To be honest about that, we use a slightly fatter bell curve (the t **distribution**) and multiply by a bit more than two:

```
round(c(n_6 = qt(0.975, df = 5),
        n_43 = qt(0.975, df = 42),
        very_large = qnorm(0.975)), 2)
```

```
      n_6      n_43 very_large
     2.57      2.02       1.96
```

The correction is trivial for the samples in most published work — and substantial for samples the size of this room.

6.6 By hand and by function

```r
m <- mean(x)
se <- sd(x) / sqrt(length(x))
margin <- qt(0.975, df = length(x) - 1) * se
lower <- m - margin
upper <- m + margin
c(lower, upper)
```

```
[1] 23.53139 32.81524
```

```r
t.test(x)$conf.int
```

```
[1] 23.53139 32.81524
attr(,"conf.level")
[1] 0.95
```

Between roughly 23.5% and 32.8%. But were these 43 countries drawn at random from all countries?

6.7 An interval only covers chance

Imai (2017, ch. 7) checks state-level polls from the 2008 US presidential election against the actual results.

- The 95% intervals contained the true result in only about **59%** of states.

That is Session 1's distinction again. **Inference handles chance; identification handles everything else.**

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- Why? The polls were **biased** — shifted in one direction. An interval centred in the wrong place misses, however well its width is computed.
- Standard errors measure one thing: **the luck of the draw**. Not bias in who answers, not measurement error, not an unfair comparison.

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Section 7

7. Hypothesis tests

7.1 A lady tasting tea

Fisher's classic test (Imai 2017, ch. 7): a colleague claims she can taste whether milk or tea went into the cup first.

- Eight cups, four of each, in random order. She must pick the four "milk first".

That is the whole logic of a test: assume nothing is going on, and ask how surprising the data would be.

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- Eight cups, four of each, in random order. She must pick the four "milk first".
- **Suppose she is just guessing.** There are 70 ways to pick four cups out of eight. Only one is all correct: a chance of 1 in 70, about 1.4%.

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- **Suppose she is just guessing.** There are 70 ways to pick four cups out of eight. Only one is all correct: a chance of 1 in 70, about 1.4%.
- She gets all eight right. Either a rare event happened, or the assumption "she is guessing" is wrong.

That is the whole logic of a test: assume nothing is going on, and ask how surprising the data would be.

7.2 The logic in five steps

1. State what "nothing going on" means: no difference, no effect, just guessing (the **null hypothesis**).

We can reject the null; we never prove it. "Not significant" means "not enough evidence", never "no effect".

7.2 The logic in five steps

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4. The p**-value**: if nothing were going on, how often would we see an estimate at least this far out?
5. If that is rare enough (by convention, under 5%), reject "nothing going on".

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7.3 How far is far?

Every test in this course asks the same question:

How many standard errors away from "nothing" is my estimate?

- Five points means nothing on its own. Five points when chance moves things by one is a lot; when chance moves things by twenty, it is nothing.

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- So a large test statistic can come from a **large effect**, a **small standard error** — or both. With enough data, trivial differences become "significant".
- Every regression coefficient in Module II comes with exactly this number.

7.4 What a p-value is not

- It is **not** the probability that the null is true. It is calculated *assuming* the null is true (Imai 2017, ch. 7).

Report the estimate and its interval. The p-value is, at best, a footnote to those.

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- It is **not** the probability that your result is a fluke.
- It is **not** a measure of how big or important an effect is.
- $p = 0.049$ and $p = 0.051$ are not different findings. 5% is a convention, and a fairly arbitrary one.

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7.5 Two ways to be wrong

	Nothing is going on	**Something is going on**
We reject "nothing"	**False alarm** (Type I error)	Correct
We do not reject	Correct	**Missed effect** (Type II error)

- You choose the false-alarm rate (the 5%). The miss rate depends on the true effect, the noise and the sample size. The chance of *not* missing a real effect is the test's **power**.

7.5 Two ways to be wrong

	Nothing is going on	**Something is going on**
We reject "nothing"	**False alarm** (Type I error)	Correct
We do not reject	Correct	**Missed effect** (Type II error)

- You choose the false-alarm rate (the 5%). The miss rate depends on the true effect, the noise and the sample size. The chance of *not* missing a real effect is the test's **power**.
- Which error is worse depends on the stakes: wrongly concluding your climbing gear is unsafe costs you a purchase; wrongly concluding it is safe can cost you your life (Dean & Illowsky 2018, ch. 9).

7.6 “Not significant” is often “not informative”

Imai (2017, ch. 7) revisits a famous experiment: pupils randomly assigned to small or regular classes (the STAR project).

- Estimated effect of small classes on reading scores: **3.5 points**. 95% interval: **from about** -1.7 **to** 8.7.

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- Not significant. Does that mean small classes do nothing? No: the data are compatible with no effect **and** with a sizeable one.
- With few cases, real effects are routinely missed — and the few small-sample findings that *do* clear 5% tend to be the exaggerated ones.

7.7 Many tests, some “findings”

- Paul the Octopus “predicted” all 8 of Germany’s 2010 World Cup matches. If he was guessing: a chance of 1 in 256 (Imai 2017, ch. 7).

An agent will happily run fifty tests in a second. Decide **what** you are testing **before** you see the results.

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- Impressive — until you remember how many animals were “predicting” that summer. Run enough tests and something will clear the bar.
- Ten tests where nothing is going on: about a **40%** chance at least one comes out “significant”.
- Journals add to it: published p-values bunch suspiciously just below 0.05 (Gerber and Malhotra 2008).

An agent will happily run fifty tests in a second. Decide **what** you are testing **before** you see the results.

7.8 Comparing two groups

The most common test in applied work: do two groups differ on average?

```
tt <- t.test(unvoting ~ icc, data = gh)
round(c(tt$estimate, p = tt$p.value), 3)
```

```
mean in group 0 mean in group 1                   p
         24.242          46.574               0.000
```

ICC members voted with the US at the UN 46.6% of the time in 2003; non-members 24.2%. A gap of about 22 points, far beyond chance. R's `t.test()` does not assume the two groups are equally spread out (the "Welch" test in Dean & Illowsky, ch. 10).

7.9 And a harder one

```
tt2 <- t.test(unvoting ~ article98, data = gh)
round(tt2$estimate)
```

```
mean in group 0 mean in group 1
             37              30
```

```
round(c(p = tt2$p.value,
        tt2$conf.int), 3)
```

```
     p
 0.039  0.377 14.059
```

States that signed an Article 98 agreement voted with the US *less* often (29.6% against 36.9%), $p = 0.039$; the interval for the gap runs from 0.4 to 14.1 points.

7.10 What would you conclude?

- The interval nearly touches zero. A handful of different cases would push p across 0.05 — so the threshold is doing no real work here.

The test is fine. The question it answers is not the one we care about. Hold on to this example: it is the problem Module IV exists to solve.

7.10 What would you conclude?

- The interval nearly touches zero. A handful of different cases would push p across 0.05 — so the threshold is doing no real work here.
- The sign runs against the obvious story: states that signed an immunity deal with the US were *less* aligned with it at the UN, not more.

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7.10 What would you conclude?

- The interval nearly touches zero. A handful of different cases would push p across 0.05 — so the threshold is doing no real work here.
- The sign runs against the obvious story: states that signed an immunity deal with the US were *less* aligned with it at the UN, not more.
- Nothing here is causal. States chose whether to sign, and the US chose whom to press. Signers and non-signers differ in every respect at once.

The test is fine. The question it answers is not the one we care about. Hold on to this example: it is the problem Module IV exists to solve.

Section 8

8. Lab: inference on real data

8.1 Lab B: specify, run, verify

To the agent

Write scripts/lab_inference.R. Using data_2011-07-26.dta: (1) average unvoting with a 95% confidence interval; (2) test whether it differs from 50; (3) compare pos1 between European and other states; (4) compare unvoting between Article 98 signers and non-signers — for all states, then separately for ICC members and non-members. Report estimate, interval and p-value each time. Run it.

Verify:

1. Rebuild one interval yourself: average $\pm$ `qt(0.975, n - 1)` $\times$ standard error. Does it match the agent's?
2. How many states are in each group? Who dropped out?
3. In (4), what happens to the gap when you split by ICC membership? Which number would you report — and is any of them an *effect*?
4. Did the agent call anything a cause? Check against `AGENTS.md`. One line in `log.md`; finish at home.

Section 9

9. For next time

9.1 Where this goes

- **Module II:** regression. Every coefficient comes with a standard error, a test statistic and an interval — today's machinery, applied to a slope instead of an average.

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- **Module II:** regression. Every coefficient comes with a standard error, a test statistic and an interval — today's machinery, applied to a slope instead of an average.
- **Module III:** panel data, where the assumption behind today's standard errors — that cases are independent of each other — breaks down.
- **Module IV:** causal inference, which asks what would license reading any of today's differences as **effects**.

9.2 For next time

Read:

- Goldsmith & Horiuchi (2012) — we replicate its OLS models in Session 3. Come with one question about its research design.
- Dean & Illowsky (2018), Chapter 12; Imai (2017), Chapter 4
- If today went fast: Dean & Illowsky (2018), Chapters 8–10; Imai (2017), Chapter 7 — the decoder on the next slide translates their notation

Do:

- Finish Lab B and add a line to `log.md`
- *Optional:* ask the agent to simulate 5,000 comparisons of two groups drawn from the **same** population. How often is $p < 0.05$? Now test five outcomes each time: how often is **any** of them significant?

Session 3: linear regression — the same questions, asked of a slope.

9.3 Decoder: notation you will meet in the readings

In the readings	What it means on today's slides
$\sigma/\sqrt{n}$, $s/\sqrt{n}$	the standard error of an average: spread of the cases, shrunk by the square root of n
$\bar{x} \pm$ EBM	the confidence interval: estimate plus or minus the margin of error ("error bound")
CL, z^*, t^*	confidence level (e.g. 95%) and the "about two" we multiply by
$df = n - 1$	degrees of freedom: how fat the t curve is; grows with the sample
H_0, H_a	"nothing going on" (null) and its alternative
α	the false-alarm rate you accept (usually 5%)
β, $1 - \beta$	the chance of missing a real effect; and power
p', $\hat{p}$	a share estimated from the sample
one-/two-sided	whether "surprising" counts in one direction or both

You do not need to manipulate these. You need to recognise them.